HaseebOS

Personal Intelligence Operating System

Complete Architecture & Functional Design

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This document covers:

Complete system architecture • Data flows • Integration design • Brain and context design • Morning briefing design • Decision and delegation engine • Memory and follow-up system • Open items monitoring • Email intelligence • ERP financial monitoring • Project status tracking • Risk register monitoring • OKR system integration

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1. System Overview & Design Principles

HaseebOS is Abdul Haseeb's personal intelligence operating system — a fully automated chief-of-staff built into Claude.ai. It captures every signal across all communication and work channels, processes them through a central AI brain, maintains a unified open items database, and surfaces a daily prioritised briefing inside Claude conversations. Abdul never opens a separate dashboard — everything happens inside this Claude project.

Core Design Principles

Principle	Description
Claude-native interface	All interaction happens inside Claude.ai conversations. Cowork runs background jobs silently.
Single open items DB	Every task, delegation, follow-up, and alert lives in one Notion database — no fragmentation.
One central brain	A unified context layer holds all knowledge: entities, commitments, decisions, rules, thought pipeline.
Five incoming feeds	Gmail, WhatsApp, Google Chat, ESS/systems, and Known (Vertex AI) all flow into one normalisation layer.
Decisions-first learning	Every decision Abdul makes is logged. Patterns reaching 90% consistency over 30 days become candidates for a
Drafts before sending	Nothing goes out automatically in Year 1. All actions are suggested, reviewed, then executed.
Entity-linked everything	All items link to a unique entity (contact, account, project) in the knowledge graph. No orphaned data.
ERP-aware	Financial figures, project status, risk registers, and OKRs are monitored from SAP/Odoo and surfaced proactively

2. Complete Architecture — Six Layers

HaseebOS is organised into six functional layers. Data flows top-to-bottom: incoming feeds are normalised and matched against the open items database, the central brain provides context for action identification, the action engine generates suggestions, and the Claude conversation is where Abdul reviews and acts. All decisions flow back into the brain.

LAYER 1 — INCOMING FEEDS (5 sources)	
Feed Source	**What Is Read**
Gmail	All unread · CC/BCC · attachments · escalations · invoices · meeting invites
WhatsApp	Drive summaries · Tasks.json · voice transcripts · client threads
Google Chat	MD Office · Sales · SAP · Cyber team spaces · direct messages
ESS / TMC Systems	Appraisal actions · Odoo CRM alerts · SAP notifications · approval requests
Known (Vertex AI)	Org-wide email intelligence · risks · opportunities · employee flags · client sentiment

LAYER 2 — OPEN ITEMS DB (Notion)		
Field	**Description**	**Source**
Item ID	Unique identifier (ITEM-NNN)	Auto-generated
Title	One-line description of the item	Feed or manual
Entity	Linked contact / account / project (from knowledge graph)	Entity resolver
Type	Task / Email / Delegation / Alert / ERP / OKR / Risk	Classifier
Status	Open / In Progress / Delegated / Blocked / Done / Overdue	Updated by system
Priority	Critical / High / Medium / Low	Rules engine
Owner	Abdul or named team member	Delegation framework
Delegatee	Person item was delegated to (if applicable)	On delegation
Due date	Deadline (if any)	From source or Abdul
Source feed	Gmail / WhatsApp / Chat / ESS / Known / ERP / OKR	Feed tag
Source ref	Gmail thread ID / Drive file / Odoo record / OKR ID	Source system
Delegation trail	Log of all handoffs with timestamp and outcome	Append-only
Last update	When item was last touched by system or Abdul	Auto-timestamp
Notes	Context, decisions, follow-up log	Append-only

LAYER 3 — CENTRAL BRAIN (Claude Project + Notion + Drive)			
Component	**What It Contains**	**Where It Lives**	**Update Frequency**
TMC Master Context	Clients, projects, team, commitments, risks, pipeline	Google Drive + Notion	Every 6 hours
Entity Knowledge Graph	Contacts, accounts, projects — uniquely identified and linked	Notion database	On every new entity detected
Decisions Log	Every decision Abdul makes — action, context, outcome, override, pattern	Google Sheets / Notion	After every session
Rules Engine	Daily rule book, delegation matrix, escalation rules, check triggers	Google Drive	Human-edited, read daily
Thought Pipeline	Strategic thinking, long-form analysis, opportunity notes, reflections	Notion pages	Auto-written + Abdul edits

Component	What It Contains	Where It Lives	Update Frequency
Claude Project Memory	Full conversation history — searchable, used as primary context	Claude.ai Project	Every conversation

LAYER 4 — ACTION IDENTIFICATION ENGINE

The action identification engine matches every incoming feed item against the open items database and the central brain. It resolves entities, detects whether an item is a progress update on an existing task or a new item entirely, classifies intent, and generates a ranked list of suggested actions for Claude to present to Abdul.

Step	What Happens	Output
1. Entity resolution	Extract person, account, or project from feed item. Match against entity knowledge graph	Entity ID linked
2. Open item matching	Check if feed item relates to an existing open item (by entity + context). Update	Item updated or new item created
3. Intent classification	Classify item: new task / progress update / escalation / information / risk / opportunity	Item type + priority
4. Brain context load	Load master context + entity history + decisions log + rules engine for this entity	Contextual background
5. Action generation	Claude API generates ranked action suggestions with full draft, delegatee, and context	Action list for steering wheel

LAYER 5 — ACTION TYPES

Action	What Happens	Tools Used	DB Update
Reply	Draft email created. Abdul reviews inline and sends from Gmail	Gmail API · Claude API	Item status → In Progress / Done
Delegate	Assignee identified. Chat message or email drafted. Delegation logged	Google Chat · Gmail · Notion	Owner updated · Trail logged
Schedule meeting	Calendar event created. Attendees added. Meet link included	Google Calendar API	Item linked to event
Close / resolve	Item marked done. Outcome noted. Entity record updated.	Notion API	Status → Done · Date logged
Update thought pipeline	Notion entry drafted. Abdul reviews and publishes	Notion API · Claude API	Thought pipeline entry created
ERP action	Flag financial item / update Odoo record / note	SAP API / Odoo API	ERP item linked to open item
OKR alert	Surface behind-target OKR. Draft update or corrective action.	OKR system API	OKR item flagged in open items
Escalate	Route to MD personally. Block delegation. Mark critical.	Gmail · Chat	Priority → Critical · Owner → Abdul

LAYER 6 — CLAUDE-NATIVE STEERING WHEEL (Claude.ai)

Component	Description
Morning briefing message	Structured daily briefing surfaced in Claude conversation at 8:30 AM PKT
Item-by-item review	Each action item shown with context, draft, and options. Abdul responds inline.
Live action execution	Gmail drafts, Calendar events, Notion updates, Chat messages — executed without leaving Claude
Auto-log to brain	Every decision automatically logged to Decisions Log and Open Items DB
Claude Tasks plugin	Evaluated for native task management — use if sufficient, Notion as fallback
Team briefers (8am PKT)	Personalised Google Chat briefers dispatched to each team space by Cowork

3. Data Flows & Integration Design

3.1 Master data flow — feed to action

Every data item in HaseebOS follows the same path: ingested from a feed source, normalised into a standard format, matched against the open items DB via entity resolution, enriched with brain context, processed through the action identification engine, surfaced to Abdul in Claude, actioned, and the outcome logged back to the brain. This cycle runs continuously — Cowork jobs run on schedule and on-demand.

1.	**Feed source fires**	Gmail / WhatsApp / Chat / ESS / Known / ERP / OKR
2.	**Cowork ingestion job**	Normalises to standard feed item format. Extracts entities, intent, urgency.
3.	**Entity resolver**	Matches person/account/project against knowledge graph. Creates new entity if unknown.
4.	**Open items matcher**	Checks if item relates to existing open item. Updates status or creates new item.
5.	**Brain context loader**	Reads master context, decisions log, entity history for this item.
6.	**Action generator (Claude)**	Generates ranked suggested actions with full drafts and delegatee pre-named.
7.	**Steering wheel (Claude conversation)**	Abdul reviews briefing, reads each item, decides action inline.
8.	**Action executor**	Gmail/Calendar/Notion/Chat APIs called to execute the chosen action.
9.	**Decision logger**	Decision, action taken, and outcome logged to Decisions Log and Open Items DB.
10.	**Brain updater**	Master context, entity graph, and thought pipeline updated with new information.

3.2 Integration architecture — system connections

HaseebOS integrates with eight external systems. Each integration has a defined read/write pattern, authentication method, and data contract.

System	Direction	Data Read	Data Written	Auth Method	Frequency
Gmail (API v1)	Read + Write	Unread messages, threads, labels, attachments	Draft replies, send emails	OAuth2 refresh token	On schedule + real-time
Google Calendar	Read + Write	Events, RSVPs, attendees, conflicts	New events, RSVP responses	OAuth2	On schedule
Google Drive	Read + Write	Context files, WhatsApp summaries, rule books	Dashboard HTML, updated context files	OAuth2	Every 6 hours
Google Tasks	Read + Write	All open tasks, task lists, due dates	New tasks, status updates	OAuth2	Every 15 min

System	Direction	Data Read	Data Written	Auth Method	Frequency
Google Chat	Write	—	Team briefers, delegation messages	Webhook + OAuth2	8am PKT daily
Notion	Read + Write	Open Items DB, entity graph, thought pipeline	New items, status updates, context pages	Notion API key	Continuous
WhatsApp (Meta API)	Read via Drive	Drive summaries and Tasks.json from Antigravity pipeline	—	Drive OAuth2	Nightly digest
SAP / Odoo (ERP)	Read + Write	Financials, project data, risks, OKRs, AR/AP	Status updates, risk flags	REST API / BAPI	Every 6 hours
Known (Vertex AI)	Read	Org risk digest, opportunities, employee flags, sentiment	—	GCP service account	Nightly digest

4. Central Brain — How It Works

The Central Brain is the intelligence core of HaseebOS. It is not a single system but a constellation of six interconnected knowledge stores that together give Claude complete context before generating any suggestion. Every morning session, and every ad-hoc query, begins with the brain loading the latest state of all six components.

4.1 TMC Master Context File

The authoritative, always-current source of truth about Abdul's world. Updated every 6 hours by a Cowork job that reads Gmail, WhatsApp summaries, Calendar, Odoo CRM, SAP data, and Known outputs. Human edits always take precedence.

Section	Contents
Active clients	Name, industry, CXO contact, last interaction, sentiment, open commitments, next action
Active projects	Name, phase, TMC owner, RAG status, key risks, next milestone
Open commitments	What was committed, to whom, by when, source, current status
Key relationships	CXO/sponsor contacts per client, last touchpoint, sentiment trend
MD Office team	Current assignments, capacity, flags, delegated items per person
Business pipeline	Opportunities from Odoo CRM: stage, value, next action, owner
Active risks	Oracle/Known-sourced risks: severity, mitigation, owner, age
Decisions this week	Summary of last 7 days of Decisions Log
Thought pipeline triggers	Items flagged for Notion entry, not yet written
ERP snapshot	Cash position, overdue AR, project cost vs budget, OKR progress

4.2 Entity Knowledge Graph

Every person, company, and project that appears in any feed or conversation is identified as a unique entity. Entities are linked to open items, decisions, and thought pipeline entries. This prevents duplication and enables the system to track a relationship across Gmail, WhatsApp, and ERP data simultaneously.

Entity Type	Key Fields	Linked To
Contact	Name, email, phone, company, role, relationship strength, sentiment	Open items, emails, WhatsApp threads, decisions
Account (Client)	Name, industry, CXO, project status, sentiment, AR balance, pipeline stage	Contacts, projects, open items, ERP records
Project	Name, client, phase, RAG, owner, budget vs actual, timeline, risks	Tasks, risk register, ERP project codes, OKRs
Opportunity	Name, client, stage, value, owner, close date, next action	Contacts, pipeline in Odoo, open items
Risk	Description, category, severity, mitigation, owner, status, source	Projects, accounts, Known digest
OKR	Objective, key results, target, actual, owner, period, status	Projects, accounts, ERP metrics

4.3 Decisions Log — learning mechanism

Every decision Abdul makes during a Claude session is automatically appended to the Decisions Log. Weekly, a Cowork analysis job calculates match rates per pattern. Patterns reaching 90% consistency over 30+ decisions are flagged as candidates for auto-action promotion — Abdul reviews and approves elevation each Monday.

Column	Values
Date / Time (PKT)	Timestamp of decision
Session type	Morning Steering Wheel / Intraday / Auto-action
Item type	Email / WhatsApp / Task / Calendar / ERP / OKR / Known / Manual
Entity	Linked entity from knowledge graph
Suggested action	What Claude recommended
Abdul's decision	Approved / Overrode / Delegated / Snoozed
Action taken	Specific execution (draft created, task assigned, etc.)
Match	TRUE = Abdul accepted · FALSE = overrode
Override reason	Why Abdul chose differently (optional text)
Outcome (30 days)	Positive / Negative / Neutral — assessed later

5. Morning Briefing Design

The morning briefing is the primary interaction between Abdul and HaseebOS. It runs every day at 8:30 AM PKT, surfaced as a structured message inside this Claude conversation. Cowork prepares all data by 7:30 AM. The briefing is conversational — Abdul responds inline and Claude executes actions immediately. There is no separate dashboard or app to open.

5.1 Briefing preparation — Cowork jobs (7:30 AM PKT)

Step	Job	What It Does
1	Feed ingestion	Reads all 5 feeds, normalises items, updates Open Items DB
2	Brain refresh	Updates master context from all sources, refreshes entity graph
3	ERP snapshot	Fetches latest financials, project status, AR/AP, OKR progress from SAP/Odoo
4	Known digest	Reads latest Vertex AI risk/opportunity/flag digest
5	Action generation	Claude API generates ranked action items with drafts for all open items
6	Briefing assembly	Compiles structured briefing message with all sections and action items
7	Team briefers	Dispatches personalised Google Chat messages to each team space at 8:00 AM PKT
8	Briefing delivery	Posts morning briefing to this Claude conversation at 8:30 AM PKT

5.2 Briefing structure — what Abdul sees

The briefing is structured as a single Claude message with clear sections. Abdul reads and responds conversationally — Claude executes each decision live.

SECTION A — Good morning summary

- Date, day, weather in Lahore/Karachi
- Open items count: Critical / High / Medium / Low
- Today's meeting count + first meeting time
- Key figure: overdue items, pending delegations awaiting update
- One-line highlight: biggest item of the day

SECTION B — Critical items (must act today)

- All Critical priority items from Open Items DB
- Each item: entity name, what happened, suggested action, full draft
- CXO escalations always appear here — never delegated
- Overdue items that have crossed 48-hour threshold
- ERP alerts: overdue AR > threshold, budget overrun, OKR critical miss

SECTION C — Today's calendar

- Real meetings only (Reclaim noise filtered out)
- Each meeting: time, attendees, context from master context
- Unresponded invitations flagged with suggested RSVP
- Meetings where Abdul needs to prepare: prep note suggested

- Tomorrow's key meetings flagged for advance awareness

SECTION D — WhatsApp digest

- Summary of all client WhatsApp conversations from last 24 hours
- Threads awaiting Abdul reply flagged with suggested response
- Any negative sentiment thread surfaced immediately
- New client contacts detected and flagged for entity creation

SECTION E — Delegation follow-up

- All delegated items with no update in 48+ hours
- Each item: who it was delegated to, when, what was asked, last status
- Suggested chasing message pre-drafted per person
- Overdue delegations escalated to High priority

SECTION F — ERP & OKR snapshot

- Cash position vs plan (from SAP/Odoo)
- Overdue accounts receivable above threshold
- Project budget variance: over 10% flagged
- OKR progress: any key result below 70% of target highlighted
- Risk register: any new High/Critical risks added

SECTION G — Known intelligence feed

- Org-wide risks surfaced by Vertex AI — employee flags, client issues
- Opportunities identified from email analysis across TMC
- Employee sentiment flags (private to Abdul only)
- Follow-up items: TMC threads with no response > 48 hours

SECTION H — Thought pipeline prompts

- Reflection prompts based on patterns observed this week
- Draft Notion entries awaiting Abdul's review and publish
- Strategic questions the brain has identified as worth considering
- This week in review summary (Fridays only)

6. Decision & Delegation Engine

The decision and delegation engine is embedded in the Claude API system prompt. For every action item, Claude determines: can this be delegated? To whom? With what brief and what channel? The engine uses the delegation matrix, entity knowledge, team capacity from the master context, and the decisions log to make accurate routing decisions.

6.1 Decision flow — item to action

Step	Question	Outcome
1. Classify	What type of item is this?	Type determined: email / task / ERP alert / risk / OKR
2. Escalation check	Is this a client CXO/MD escalation?	YES → Abdul personally, never delegate, mark Critical
3. Privacy check	Is this HR / increment / performance?	YES → Abdul personally, never in team briefers
4. Financial check	Is this a financial approval (invoice / payment)?	YES → Abdul approves, Salman Sohail executes
5. Delegation check	Can this be delegated? Who is best placed?	Delegation matrix consulted → assignee named
6. Capacity check	Is the proposed assignee at capacity?	YES → propose alternative or defer to Abdul
7. Draft generation	What should be said / done?	Full draft created: email / Chat message / task brief
8. Abdul decides	Approve / Override / Delegate / Snooze?	Decision executed live in Claude conversation
9. Log decision	Record what happened and whether it matched suggestion	Decisions Log updated, Open Items DB updated

6.2 Delegation matrix

Item Type	Default Route	Primary Assignee	Channel	Abdul CC?
Client CXO / MD escalation	Abdul personally	—	Gmail (personal)	N/A
Client sentiment negative	Abdul personally	—	Gmail	N/A
Invoice / payment approval	Abdul approves → Salman executes	salman.sohail@tmcltd.ai	Gmail + approval workflow	Yes
HR / increment / performance	Abdul personally	—	Private	N/A
Technical SAP query (pre-sales)	Delegate	mohsin@tmcltd.com	Google Chat	No
Meeting scheduling	Delegate to coordinator	madeeha.rahat@tmcltd.ai	Google Chat	No
Internal team follow-up	Delegate to team lead	Relevant lead	Google Chat	No
Business development follow-up	Delegate	saba.naeem@tmcltd.ai	Gmail / Chat	No
Partnership / vendor queries	Delegate + Abdul CC'd	Relevant lead	Gmail	Yes
ERP financial alert	Abdul reviews, Salman actions	salman.sohail@tmcltd.ai	Chat + Odoo task	Yes
Project risk flag	Delegate to project owner	Relevant PM	Chat + risk register	Yes
OKR behind target	Abdul reviews, owner notified	OKR owner	Chat + OKR system	Yes
Media / press enquiries	Abdul personally	—	Gmail (personal)	N/A

Item Type	Default Route	Primary Assignee	Channel	Abdul CC?
Client CXO / MD escalation	Abdul personally	—	Gmail (personal)	N/A
Client sentiment negative	Abdul personally	—	Gmail	N/A

Item Type	Default Route	Primary Assignee	Channel	Abdul CC?
Invoice / payment approval	Abdul approves → Salman executes	salman.sohail@tmcltd.ai	Gmail + approval workflow	Yes
HR / increment / performance	Abdul personally	—	Private	N/A
Technical SAP query (pre-sales)	Delegate	mohsin@tmcltd.com	Google Chat	No
Meeting scheduling	Delegate to coordinator	madeeha.rahat@tmcltd.ai	Google Chat	No
Internal team follow-up	Delegate to team lead	Relevant lead	Google Chat	No
Business development follow-up	Delegate	saba.naeem@tmcltd.ai	Gmail / Chat	No
Partnership / vendor queries	Delegate + Abdul CC'd	Relevant lead	Gmail	Yes
ERP financial alert	Abdul reviews, Salman actions	salman.sohail@tmcltd.ai	Chat + Odoo task	Yes
Project risk flag	Delegate to project owner	Relevant PM	Chat + risk register	Yes
OKR behind target	Abdul reviews, owner notified	OKR owner	Chat + OKR system	Yes
Media / press enquiries	Abdul personally	—	Gmail (personal)	N/A

7. Memory, Learning & Pattern Retention

HaseebOS is designed to get smarter over time. Three mechanisms drive this: the Decisions Log captures every choice Abdul makes; a weekly analysis job identifies patterns; and the Claude Project memory stores the full conversation history as searchable institutional knowledge.

7.1 What is remembered and how

What	Where Stored	How Used	Retention
Every decision Abdul makes	Decisions Log (Notion/Sheets)	Pattern analysis · calibration of suggestions	Permanent
All Claude conversations	Claude Project memory	Searched before every session · primary context	Project lifetime
Entity history	Entity knowledge graph (Notion)	Context on any contact or account · relationship history	Permanent
Draft outcomes	Decisions Log	Tracks whether drafts were used as-is or modified	Permanent
Delegation outcomes	Open Items DB + Decisions Log	Tracks whether delegations succeeded · assignee calibration	Permanent
Override reasons	Decisions Log	Why Abdul chose differently — used to improve future suggestions	Permanent
ERP snapshots	Master context (Drive)	Financial trend tracking · budget variance history	Rolling 12 months
OKR progress history	Master context + OKR system	Trend analysis · pattern of slippage	Per OKR period
Thought pipeline	Notion pages	Strategic memory · reference for future decisions	Permanent

7.2 Pattern learning — path to auto-action

The learning engine runs every Sunday at 6:00 AM PKT. It analyses the Decisions Log for the past 30+ days, calculating match rates per item type, entity category, and action type. Patterns reaching 90% consistency are presented to Abdul as promotion candidates on Monday morning.

Days 1-30	**Full human review**	Every suggestion reviewed. All decisions logged.
Day 31+	**Pattern analysis**	Weekly job identifies patterns with 90%+ match rate over 30+ decisions.
Monday briefing	**Promotion review**	Candidates presented to Abdul: 'This pattern has been consistent — approve auto-action?'
Post-approval	**Supervised auto-action**	Actions execute automatically but remain visible in briefing as 'auto-actioned' log.
Month 6+	**Full auto-action**	High-confidence patterns run silently. Monthly audit reviews auto-action log.

8. Delegation Follow-Up & Task Monitoring

HaseebOS never loses a delegated item. Every delegation is logged in the Open Items DB with full trail. A dedicated follow-up engine monitors all open and delegated items continuously, surfacing any that have gone stale, missed their deadline, or produced an update that needs Abdul's attention.

8.1 Delegation lifecycle

Phase	What Happens	System Action
Created	Abdul delegates via Claude conversation	Item created in Open Items DB: status=Delegated, owner=assignee, delegatio
Notified	Assignee receives Google Chat or Gmail message with full brief	Message sent via Chat/Gmail API. Delivery confirmed logged.
In progress	Assignee works on item. System monitors for updates.	Incoming feeds scanned for references to this item or entity.
Update detected	Email / Chat message from assignee references the item	Entity resolution links message to open item. Status updated. Abdul notified.
48-hour check	No update received in 48 hours	Item flagged in next morning briefing. Chasing message drafted.
Overdue	Item has passed due date with no resolution	Priority escalated to High. Item moved to Critical section of briefing.
Escalation	Overdue + 72 hours OR Abdul flagged Critical	Item routed back to Abdul. Cannot be re-delegated without Abdul approval.
Closed	Assignee reports completion / Abdul marks done	Status=Done. Outcome noted. Delegation trail completed with result.

8.2 Open items monitoring — critical item tracking

The system continuously monitors open items for status changes, approaching deadlines, and critical conditions. Items are re-prioritised dynamically as new information arrives.

Condition	Trigger	Action
Item age > 48 hours, no update	Rolling check every 6 hours	Flag in briefing. Chase message drafted.
Item age > 72 hours, no update	Rolling check	Priority upgraded to High. Escalation path suggested.
Due date within 24 hours	Scheduled check	Item moved to Critical section. Reminder drafted.
Item is past due date	Scheduled check	Priority → Critical. Abdul must act or re-delegate.
Delegatee capacity full	On new delegation creation	Warning surfaced before delegation is confirmed.
Entity sentiment = Negative	On WhatsApp/Known feed update	All open items for that entity upgraded to High priority.
ERP: budget overrun > 10%	On ERP snapshot update	Project item flagged. Owner notified. CFO CC suggested.
OKR: key result below 70%	On OKR system sync	OKR item surfaced in briefing. Corrective action drafted.
Known: Critical risk for entity	On Known feed digest	All open items for entity reviewed. Abdul alerted immediately.

9. Email Intelligence — Unread, Critical & Escalated

The email intelligence layer reads all unread Gmail from the last 24 hours and applies a multi-stage classification pipeline before surfacing any items to Abdul. The goal is that Abdul sees only what requires his specific attention — everything else is either auto-handled, delegated, or acknowledged as FYI.

9.1 Email classification pipeline

Stage	Classification	Criteria	Action
1	CRITICAL	From client CXO/CIO · Legal or compliance · Board-level · Media	Surface immediately at top of briefing. Abdul handles personally.
2	ESCALATION	Reply chain > 72 hours with no TMC response · Client frustration detected	Surface in Escalation section. Draft apology + resolution.
3	FINANCIAL	'invoice', 'payment', 'overdue', 'statement' in subject · From SAP billing	Route to Salman Sohail. Abdul CC'd. Odoo record flagged.
4	APPROVAL	Requires Abdul's explicit sign-off (HR, contract, purchase)	Surface with context. Approve/reject inline. Log decision.
5	DELEGATION	Can be handled by team member based on subject/sender	Delegation matrix applied. Draft forwarding message. Abdul confirms.
6	FYI	Information only. No action required. No reply needed.	Logged to master context if entity-relevant. Marked read. Not surfaced
7	NOISE	Newsletters, automated notifications, marketing, system alerts	Archived automatically. Not surfaced. Log count only.

9.2 Known-powered email intelligence (org-wide)

The Known (Vertex AI) system reads every email sent and received across the entire TMC organisation via Google Workspace domain-wide delegation. It does not surface individual email content to Abdul — it surfaces derived intelligence: risks, opportunities, follow-up gaps, and employee signals.

Output Type	What It Detects	How Surfaced to Abdul
Client risk	Client expressing frustration, delay, or complaint in email to any TMC person	CRITICAL or HIGH in Section G of briefing
Opportunity	Client or prospect mentioning new project, budget, or expansion in email	Opportunity entry in briefing. Odoo CRM draft created.
Follow-up gap	Any TMC person email thread with client unanswered > 48 hours	Delegation follow-up item created. Responsible person alerted.
Employee signal	High stress, disengagement, conflict, or compliance concern detected in employee emails	Private to Abdul only — never in team briefers. 1:1 suggested.
Client sentiment trend	Rolling 7-day sentiment analysis per client based on email tone	Sentiment score in master context. Trend shown in briefing.
Risk intelligence	Legal, compliance, regulatory, or financial risk mentioned in any email	Added to risk register. Surfaced in briefing if High/Critical.

10. ERP Integration — Financial Monitoring

HaseebOS integrates with SAP and Odoo to surface financial intelligence proactively. Rather than requiring Abdul to log into ERP systems, key financial figures are fetched every 6 hours and included in the morning briefing. Anomalies and threshold breaches are automatically flagged as open items.

10.1 Financial data fetched from ERP

Data Category	Specific Metrics Fetched	Frequency	Alert Threshold
Cash & liquidity	Cash position vs plan, bank balances per entity, upcoming payments	Every 6 hours	Below 30-day runway
Accounts receivable	Total overdue AR, by client, by aging bucket (30/60/90 days)	Every 6 hours	Any invoice > 60 days
Accounts payable	Upcoming payments due, critical vendor payables, overdue AP	Daily	Overdue > 30 days
Project financials	Revenue recognised, costs incurred, budget vs actual per project	Every 6 hours	Variance > 10%
Revenue tracking	Monthly/quarterly revenue vs plan, pipeline conversion, invoicing lag	Daily	Below 85% of plan
Payroll & HR costs	Payroll run status, headcount vs budget, increment impact	Weekly (or on run)	Variance > 5%
Client billing	Unbilled deliverables, milestone billing status, retainer utilisation	Every 6 hours	Unbilled > 30 days
Purchase orders	Open POs, approval status, vendor payment terms	Daily	Unapproved > 72 hours

10.2 ERP alert routing

Alert Type	Suggested Action	Routed To
Invoice overdue > 60 days	Draft client payment reminder email	Abdul (if client is CXO) or Sofia/Sales lead
Budget overrun > 10%	Flag to project owner and CFO. Draft status note.	Project owner + Abdul CC
Revenue below 85% of plan	Surface in briefing. Review pipeline in Odoo.	Abdul personally
Payroll run upcoming	Confirm headcount and approve run	Abdul → Salman Sohail
Unapproved PO > 72 hours	Approve or reject in Odoo	Abdul (if above threshold) or Mohsin
Client billing gap > 30 days	Draft billing reminder to project lead	Project lead + Abdul CC
Cash below runway threshold	CRITICAL flag. Immediate Abdul attention.	Abdul personally — never delegated

11. Project Status & Risk Register Monitoring

HaseebOS monitors all active TMC projects continuously — pulling status from SAP project management modules, Odoo, and email/WhatsApp intelligence. RAG status is maintained in the master context and updated automatically. Risk registers are read from SAP/Odoo and enriched with Known intelligence.

11.1 Project status monitoring

Monitoring Area	Data Source	What Is Tracked	Alert Condition
RAG status	SAP PM + Email/WhatsApp signals	Red/Amber/Green per project	Any project turning Red
Milestone adherence	SAP project plan + Calendar	Planned vs actual milestone dates	Milestone missed or at risk (< 7 days to deadline)
Resource utilisation	SAP HR module + Odoo	Allocated vs actual hours per project	Utilisation < 70% or > 110%
Budget vs actual	SAP CO / PS module	Cost incurred vs approved budget	Variance > 10% triggers alert
Client satisfaction signals	Email + WhatsApp (Known + Antigravity)	Sentiment trend for each project client	Negative sentiment sustained > 3 days
Scope change requests	Email detection (Known)	Mentions of scope, change request, CR	Any scope change detected → flag to Abdul
Delivery quality flags	Email + Chat (Known)	Client complaints, re-work mentions, escalations	Any quality flag → Critical alert
Invoice milestone	SAP billing plan + Odoo	Billing milestone triggered but not invoiced	Unbilled milestone > 14 days

11.2 Risk register integration

The risk register lives in SAP/Odoo and is enriched by Known intelligence. HaseebOS reads the register every 6 hours and surfaces any new or escalated risks in the morning briefing. All risks are linked to their source entity in the knowledge graph and tracked until closed.

Risk Source	How Detected	How Surfaced
SAP/Odoo risk register	Direct API read every 6 hours	New/escalated risks in ERP snapshot section of briefing
Known (Vertex AI)	Nightly org email analysis	Risks surfaced in Known intelligence section
Email / WhatsApp signals	Keyword and sentiment detection by Antigravity/Claude	Risk item created in Open Items DB, linked to entity
Project milestone slippage	SAP project plan comparison	Risk flagged on affected project entity
Financial anomaly	ERP threshold breach	Risk created and linked to financial entity
Employee signal	Known sentiment analysis	Private risk item — visible to Abdul only

12. OKR System Integration

HaseebOS monitors OKR progress continuously and surfaces underperforming key results before they become critical. OKRs are linked to the entities (projects, teams, accounts) they relate to, enabling the system to correlate OKR slippage with root causes visible in email, project, or financial data.

12.1 OKR data model in HaseebOS

Field	Description
OKR ID	Unique identifier from OKR system
Objective	What is being achieved
Key results (list)	Measurable outcomes with target values
Owner	Person or team responsible — linked to entity graph
Period	Quarter / half-year / annual
Current value	Latest actual value (fetched from OKR system or ERP)
Target value	Agreed target
Progress %	Current / Target — calculated by system
Status	On Track (>85%) / At Risk (70-85%) / Behind (<70%) / Critical (<50%)
Last updated	When system last fetched latest value
Root cause signals	Linked email/project/financial signals that may explain slippage
Corrective action	Suggested action generated by Claude based on context

12.2 OKR monitoring rules

Condition	Action Triggered	Urgency
Key result below 50% with >25% of period remaining	CRITICAL flag. Immediate briefing. Abdul must intervene.	Critical
Key result below 70% with >10% of period remaining	Surfaced in briefing. Owner alerted via Chat. Corrective action drafted.	High
Key result between 70-85%	Flagged as At Risk. Noted in briefing. No immediate action required.	Medium
No OKR update in 14 days	Stale data warning. Request update from OKR owner.	Medium
OKR owner entity has risk flags	Correlate: OKR slippage linked to client/project issue in briefing.	High
OKR period ends in < 30 days	Deadline warning. Progress check. Final push actions drafted.	High
OKR consistently below target for 3+ weeks	Pattern flagged. Strategic review prompt added to thought pipeline.	High

13. Infrastructure & Deployment

13.1 GCP infrastructure (existing project: direct-archery-492112-m2)

Service	Purpose	Status
Cloud Run: wa-receiver	WhatsApp Meta webhook receiver — already live	LIVE
Cloud Run: haseebos-steering-wheel	Main Cowork job service (Phases 1-4 of current spec)	BUILD
Cloud Run: haseebos-ingestion	Feed ingestion job — reads all 5 feeds and updates Open Items	BUILD
Cloud Run: haseebos-brain-refresh	Context refresh — updates master context every 6 hours	BUILD
Cloud Run: haseebos-erp-sync	ERP data fetch from SAP/Odoo — financial, project, risk, OKR	BUILD
Cloud Scheduler	Cron jobs for all scheduled Cloud Run invocations	BUILD
Secret Manager	All API keys, tokens, credentials stored securely	BUILD
Vertex AI (Known)	Org-wide email intelligence pipeline — separate team building	WIP

13.2 Storage architecture

Store	What Lives There	Access Pattern
Notion (Open Items DB)	All open items, entity graph, thought pipeline, master context mirror	Read/Write via Notion API
Google Drive (HaseebOS/)	Context files, rule book, WhatsApp summaries, dashboard files	Read/Write via Drive API v3
Google Drive (WhatsApp_Knowledge/)	WhatsApp summaries and Tasks.json from Antigravity pipeline	Read-only by non-WhatsApp services
Google Sheets	Decisions Log — append-only, analysed weekly	Append via Sheets API
Claude Project (this project)	Full conversation history — primary memory source	Read via conversation search tools
GCP Secret Manager	ANTHROPIC_API_KEY, Gmail tokens, Notion key, ERP credentials	Mount as secrets in Cloud Run
SAP / Odoo	Financial records, project data, risk register, OKRs	Read/Write via REST API

14. Build Roadmap & Status

HaseebOS is built in three phases. Phase 1 (Cowork) covers the Steering Wheel and WhatsApp integration. Phase 2 covers ERP, OKR, and full brain components. Phase 3 covers Known integration and auto-action promotion.

Component	Phase	Owner	Status	Priority
WhatsApp pipeline (wa-receiver + Antigravity)	0 (pre-work)	Done	LIVE	—
GMail + Calendar + Tasks integration	1	Cowork	BUILD	P1
Steering Wheel morning job (Cloud Run)	1	Cowork	BUILD	P1
Open Items DB (Notion)	1	Cowork	BUILD	P1
Decisions Log write-back	1	Cowork	BUILD	P1
Team briefers (Google Chat)	1	Cowork	BUILD	P1
TMC Master Context refresh job	1	Cowork	BUILD	P1
Claude conversation interface (this project)	1	Native	LIVE	—
Entity knowledge graph (Notion)	2	Cowork	BUILD	P2
ERP integration (SAP / Odoo)	2	Cowork	BUILD	P2
OKR system integration	2	Cowork	BUILD	P2
Risk register monitoring	2	Cowork	BUILD	P2
Google Chat feed reader	2	Cowork	BUILD	P2
ESS / TMC systems feed	2	Cowork	BUILD	P2
Known (Vertex AI) feed integration	3	Separate team	WIP	P3
Pattern learning engine	3	Cowork	BUILD	P3
Auto-action promotion mechanism	3	Cowork	BUILD	P3